

PAPER_467 — NGC 1300 Barred Spiral Galaxy: MUGE UQFF Bar-Driven Gas Funneling, Density Waves, and Star Formation

Whitepaper Series: Star-Magic UQFF Phase 2 — Barred Spiral Galaxy Dynamics **Session:** 120 (C++ module encoded) / Whitepapers created Session 122 **Source:** grok_share_dc707f5d3.txt (Doc 64 — NGC1300UQFFModule, “MUGE Barred Spiral Galaxy NGC 1300”) **Classification:** FIRST MUGE UQFF for NGC 1300; FIRST bar-driven gas funneling term in UQFF gravity; FIRST spiral arm density wave coupling via F_{env} **Author:** Daniel T. Murphy **CP4 Class:** Pending (dc707f5d3 batch) **C++ Module:** NGC1300UQFFModule.h / NGC1300UQFFModule.cpp

Abstract

NGC 1300 is the archetypal grand-design barred spiral galaxy, with a central stellar bar that drives gas inflow along bar dust lanes toward the nucleus, and prominent two-armed spiral structure. This paper presents the complete MUGE + UQFF gravitational model for NGC 1300, incorporating bar-driven gas funneling via F_{env} (F_{bar}), spiral arm density wave pressure ($v_{\text{arm}} = 200$ km/s), $\text{SFR} = 1$ M /yr mass growth, and dark matter halo. Result: $g_{\text{NGC1300}} = 2 \times 10^3$ m/s² at $t = 1$ Gyr (environmental/fluid dominant; repulsive U_{g2} and Λ terms advance framework). The model captures NGC 1300’s iconic bar morphology as a gravitational compression through the U_{g3} external bar term.

2. Core Physics — PAPER_467

2.1 System Parameters

Parameter	Value	Notes
M (total)	1×10^{11} M ($\sim 1.989 \times 10^{31}$ kg)	Galaxy mass
r	11.79 kpc ($\sim 3.64 \times 10^{20}$ m)	Bar + spiral effective radius
SFR	1 M /yr	Star formation rate
v_{arm}	200 km/s (2×10^5 m/s)	Spiral arm density wave velocity
B	1×10^{-10} T	Galactic magnetic field
z	0.005	Eridanus supercluster distance
M_DM	$\sim 0.85 \times M$	Dark matter fraction
ρ_{fluid}	1×10^{-25} kg/m ³	Interstellar gas density

2.2 Bar-Modified Gravitational Equation

$$g_{\text{NGC1300}}(r, t) = \frac{GM_{\text{sf}}(t)}{r(t)^2} (1 + H_z t) \left(1 - \frac{B}{B_{\text{crit}}} \right) (1 + F_{\text{env}}(t)) \\ + \sum U_{gi} + \frac{\Lambda c^2}{3} + U_i + g_{\text{quantum}} + g_{\text{fluid}} + g_{\text{DM}}$$

2.3 Bar-Driven $F_{\text{env}}(t)$

The environmental force combines bar gas funneling and spiral arm density wave:

$$F_{\text{bar}} = \frac{GM_{\text{bar}}}{r_{\text{bar}}^2}$$

$$F_{\text{wave}} = k_{\text{wave}} \cdot \frac{v_{\text{arm}}^2}{r}$$

$$F_{\text{env}}(t) = F_{\text{bar}} + F_{\text{wave}}$$

Where: - M_{bar} = mass of the stellar bar ($\sim 10\%$ of M_{total}) - r_{bar} = bar half-length - $v_{\text{arm}} = 2 \times 10^5$ m/s = spiral arm pattern speed - k_{wave} = wave pressure coupling constant

Physical interpretation: The bar channels gas from 10 kpc toward the nucleus at bar fraction velocity, creating a directed gravitational compression modeled as an additional F_{env} term — the **first UQFF bar gravity compression**.

2.4 Ug Sub-terms for Barred Spiral

- **Ug1** (magnetic dipole): $U_{g1} = \mu_{\text{dipole}} \cdot B$
- **Ug2** (superconductor): $U_{g2} = B_{\text{super}}^2 / (2\mu_0)$, repulsive
- **Ug3** (external bar): $U_{g3} = GM_{\text{bar}} / r_{\text{bar}}^2$ — models bar gravity on disk
- **Ug4** (reactive energy): $U_{g4} = k_4 \cdot 10^{46} e^{-0.0005t}$

2.5 Spiral Arm Resonance Mode

When `mode = "resonance"`, the spiral arm oscillation is added:

$$g_{\text{res}}(t) = \frac{2\pi}{13.8} \cdot A \cdot \text{Re}[e^{i(\omega t + \phi)}] \cdot \cos(\omega_{\text{arm}} t)$$

This models the standing density wave pattern in NGC 1300's two-arm spiral as a quantum-scale resonant gravitational perturbation.

3. Equation Summary

$$g_{\text{NGC1300}}(r, t) = \frac{GM_{\text{sf}}(t)}{r(t)^2} (1 + H_z t) (1 - B/B_{\text{crit}}) (1 + F_{\text{bar}} + F_{\text{wave}}) + \sum U_{gi} + \frac{\Lambda c^2}{3} + U_i + g_{\text{quantum}} + g_{\text{fluid}} + g_{\text{DM}}$$

Computed Result: $g_{\text{NGC1300}} \approx 2 \times 10^{36} \text{ m/s}^2$ — env/fluid dominant; repulsive Ug2 and Λ terms establish equilibrium in the bar-spiral transition region.

4. Physical Interpretation

- **Bar-spiral coexistence:** F_{bar} channels gas inward (attractive compression), while F_{wave} provides outward spiral momentum — the balance between these in F_{env} captures the NGC 1300 morphology in gravitational terms.
 - **No AGN terms:** Unlike NGC 1316 or M51, NGC 1300 has no strong nuclear activity — the Ug4 reactive term remains cosmologically seeded rather than AGN-driven.
 - **Density wave resonance:** $v_{\text{arm}} = 200$ km/s corresponds to the co-rotation radius where stars and arms rotate at the same angular velocity — a classic barred spiral resonance reproduced in UQFF's F_{wave} term.
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5. C++ Module Reference

Module: NGC1300UQFFModule (root-level, Session 120 from grok_share_dc707f5d3.txt) **Key method:** `computeG(double t)` — returns total g_{NGC1300} in m/s^2 **Unique feature:** `computeUg3prime(double t)` — bar gravitational pull; resonance mode support **Integration point:** MAIN_1_CoAnQi.cpp barred spiral validation

QS=5 — Full UQFF integration: bar F_{env} , Ug1-Ug4, density wave resonance, DM/fluid terms.
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